IND-ACC-16 Arista Access Node QoS Congestion Event SOP (ALARM_1033)

Node: IND-ACC-16 | **Vendor:** Arista | **Severity:** Major | **Alarm Code:** ALARM_1033

Description

This Standard Operating Procedure (SOP) addresses a major alarm on the Arista Access node IND-ACC-16 at the Indiranagar Exchange site. The alarm, ALARM_1033, indicates a QoS congestion event due to microbursts, interface tail drops, and/or QoS queue starvation. This failure domain occurs on the access role due to the inherent nature of the Arista EOS (Extensible Operating System) architecture, which prioritizes traffic based on the configured QoS policies. In this scenario, the QoS policies may not be adequately configured to handle the traffic load, leading to congestion and subsequent alarm generation. The Arista 7250X-32S switch's hardware capabilities, such as the TCAM (Ternary Content-Addressable Memory) and the QoS forwarding engine, also play a role in the congestion event. The TCAM is responsible for storing the QoS tables, and any errors or inconsistencies in these tables can contribute to the congestion. Furthermore, the QoS forwarding engine's inability to handle the traffic load can also lead to congestion.

Pre-requisite Checks

- Verify that the Indiranagar Exchange site is accessible and that the remote hands are dispatched to assist with the maintenance.

- Check the status of the redundant link(s) to ensure that they are functioning properly.

- Gather all relevant information regarding the alarm, including the time of occurrence, the number of affected ports, and any previous maintenance activities performed on the node.

Potential Root Causes

Cause	Probability
Inadequate QoS policy configuration	High
TCAM errors or inconsistencies	Medium
QoS forwarding engine overload	Low

Diagnostic Commands & Expected Outputs

```
Command: show qos policy all, Expected Output: Display all configured QoS policies, Description: This command will display all configured QoS policies, including the policy names, priorities, and matching criteria.

Command: show qos counters, Expected Output: Display QoS counters, Description: This command will display the QoS counters, including the number of packets and bytes transmitted and dropped by each QoS queue.
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```
Command: show tcam summary, Expected Output: Display TCAM summary, Description: This command will display a summary of the TCAM, including the number of entries and the amount of free space.

Command: show qos forwarding statistics, Expected Output: Display QoS forwarding statistics, Description: This command will display statistics related to the QoS forwarding engine, including the number of packets and bytes processed and dropped.
```

Resolution Steps

Step 1: Step: Verify QoS policy configuration, Command: show qos policy all, Description: Verify that the QoS policy configuration is adequate and that the policies are correctly applied to the affected ports.

Step 2: Step: Adjust QoS policy configuration, Command: config qos policy priority , Description: Adjust the priority of the QoS policy to ensure that it is correctly ordered and can handle the traffic load.

Step 3: Step: Clear TCAM errors, Command: clear tcam errors, Description: Clear any TCAM errors or inconsistencies that may be contributing to the congestion event.

Step 4: Step: Verify QoS forwarding engine configuration, Command: show qos forwarding statistics, Description: Verify that the QoS forwarding engine configuration is adequate and that it can handle the traffic load.

Post-Check Verification

```
Command: show qos policy all, Expected Output: Display all configured QoS policies, Description: Verify that the QoS policy configuration has been corrected and that the policies are correctly applied to the affected ports.

Command: show qos counters, Expected Output: Display QoS counters, Description: Verify that the QoS counters have returned to normal and that the congestion event has been resolved.

Command: show tcam summary, Expected Output: Display TCAM summary, Description: Verify that the TCAM summary has returned to normal and that there are no errors or inconsistencies.
```

Roll-back Procedures

- Step: Restore original QoS policy configuration, Command: config qos policy priority , Description: Restore the original QoS policy configuration if the corrected configuration does not resolve the congestion event.

- Step: Clear TCAM errors, Command: clear tcam errors, Description: Clear any TCAM errors or inconsistencies that may have occurred during the resolution process.